KOUROVKA PROBLEM 16.21 OVER THE FIELD WITH TWO ELEMENTS

ADITYAVG13

Abstract. L. Vaserstein asked (Kourovka Notebook, Problem 16.21) whether, for $n > 2$ and a non-central matrix $\alpha \in \mathrm{SL}_n(F)$, every non-central matrix in $\mathrm{SL}_n(F)$ is a product of n matrices, each similar to α. We answer the question affirmatively for $F = \mathbb{F}_2$, and show that n may be replaced by any $k \geq n$. The proof assumes two published theorems, of Knüppel and Nielsen and of Bünger and Nielsen. It is an induction on n that sorts matrices by one property: whether they fix a vector together with a complementary hyperplane. Matrices that do reduce to dimensions $n-1$ and $n-2$ through an explicit construction. Matrices that do not have minimum rank at least 3, and for them the two cited theorems give a product of at most $2\lceil (n-1)/3 \rceil + 1$ conjugates, about $2n/3$. The number n is needed only because of matrices close to the identity, such as transvections. We give the exact number of conjugates needed for every conjugacy class of $\mathrm{GL}_3(\mathbb{F}_2)$ and $\mathrm{GL}_4(\mathbb{F}_2)$. Everything except the two cited theorems is formalized in Lean 4 with Mathlib. The work was produced almost entirely by an AI agent under the author's direction, as a test of a new computational mathematics project, and we describe how.

1. Introduction

1.1. **The question.** Problem 16.21 of the Kourovka Notebook [5], posed by L. Vaserstein, reads:

> Given a non-central matrix $\alpha \in \mathrm{SL}_n(F)$ over a field F for $n > 2$, is it true that every non-central matrix in $\mathrm{SL}_n(F)$ is a product of n matrices, each similar to α?

The 21st edition of the Notebook (2026) lists the problem as open.

The question belongs to a long line of work on products of conjugacy classes in groups; see the volume of Arad and Herzog [1] for the classical theory. For a conjugacy class C of a group G, write C^k for the set of products of k elements of C. A typical question asks how large k must be for C^k to contain all of G, or all elements outside the centre. For linear groups this is a question about matrices: given one matrix α, how many matrices similar to α does it take to write an arbitrary matrix as their product?

The number n in Vaserstein's question is natural. If α is a transvection, every matrix c similar to α has $\mathrm{rank}(c-1) = 1$. From $xy - 1 = x(y-1) + (x-1)$ we get $\mathrm{rank}(xy-1) \leq \mathrm{rank}(x-1) + \mathrm{rank}(y-1)$, so a product of k transvections g has $\mathrm{rank}(g-1) \leq k$. A matrix g with $g-1$ invertible therefore needs at least n transvections. So n cannot be lowered in general, and the question asks whether it is always enough.

1.2. **What was known.** For fields with at least four elements, Lev [6] showed that a product of two prescribed cyclic similarity classes of $\mathrm{GL}_n(F)$ contains every nonscalar matrix of the right determinant, under a condition on eigenvalues. Knüppel [3] studied the least k such that every element of $\mathrm{SL}(V)$ is a product of at most k elements of a given class. Knüppel and Nielsen [4] determined the *extended covering number* of $\mathrm{SL}_n(K)$, the least k such that $1 \in C_1 \cdots C_k$ for all non-central classes $C_1, \ldots, C_k$; it equals $n+1$ "provided n and the field

are not too small". The field $\mathbb{F}_2$ is the smallest field, and parts of their argument do not apply to it.

We do not know whether these results settle Problem 16.21 for any field, and we make no claim about fields other than $\mathbb{F}_2$.

1.3. **Results.** Over $\mathbb{F}_2$ every invertible matrix has determinant 1, so $\mathrm{SL}_n(\mathbb{F}_2) = \mathrm{GL}_n(\mathbb{F}_2)$, the centre is trivial, and a matrix similar to α is the same as a conjugate of α in $\mathrm{GL}_n(\mathbb{F}_2)$. The question becomes: is every nonidentity element of $\mathrm{GL}_n(\mathbb{F}_2)$ a product of n conjugates of any given nonidentity element?

Theorem 1. *Assume Theorems A and B below. Let $n \geq 3$, let $\alpha, g \in \mathrm{GL}_n(\mathbb{F}_2)$ with $\alpha \neq 1$ and $g \neq 1$, and let $k \geq n$. Then g is a product of k conjugates of α. In particular, Problem 16.21 has a positive answer for $F = \mathbb{F}_2$.*

The proof shows more for matrices that are far from the identity. Say that $h \in \mathrm{GL}(V)$ has a *J1 block* if it fixes a vector v and a linear form φ with $\varphi(v) = 1$; then h is the direct sum of the identity on $\langle v \rangle$ and a map on the hyperplane $\ker \varphi$. In terms of the Jordan form, h has a Jordan block of size 1 for the eigenvalue 1.

Theorem 2. *Assume Theorems A and B. Let $n \geq 5$ and let $\alpha \in \mathrm{GL}_n(\mathbb{F}_2)$, $\alpha \neq 1$, have no J1 block. Then every nonidentity element of $\mathrm{GL}_n(\mathbb{F}_2)$ is a product of k conjugates of α for every $k \geq 2\lceil (n-1)/3 \rceil + 1$.*

So for most matrices about $2n/3$ conjugates suffice, and n is forced only by matrices that split off an identity block. For $n = 3$ and $n = 4$ we computed the exact numbers (Section 4, Table 1). Most classes need only 2 conjugates, and the classes that need n are the transvections and, for $n = 4$, one class of elements of order 3.

1.4. **The cited theorems.** A matrix $A \in K^{n \times n}$ is *cyclic* if there is a vector v such that $v, Av, \ldots, A^{n-1}v$ is a basis of K^n. The *minimum rank* of A is $\delta(A) = \min_{\lambda \in \overline{K}} \operatorname{rank}(A - \lambda)$, where $\overline{K}$ is an algebraic closure of K. Knüppel and Nielsen define $\delta(A)$ as n minus the number of invariant divisors of A different from 1, and note in the proof of their Lemma 5.1 that the two definitions agree. Both notions depend only on the similarity class.

Theorem A (Knüppel and Nielsen [4, Corollary 6.9], case $K = \mathbb{F}_2$)**.** *Let $n \geq 3$, $n \neq 4$, and let $C_1, \ldots, C_m$ be non-central conjugacy classes of $\mathrm{SL}_n(\mathbb{F}_2)$ with $\delta(C_1) + \cdots + \delta(C_m) \geq n - 1$. Then $C_1 C_2 \cdots C_m$ contains a cyclic matrix.*

Theorem B (Bünger and Nielsen [2, Theorem 1.2], case $K = \mathbb{F}_2$)**.** *Let $n \geq 3$, let $M \in \mathrm{GL}_n(\mathbb{F}_2)$ be a matrix that is not a homothety, let X be a cyclic similarity class of $\mathrm{GL}_n(\mathbb{F}_2)$, and let $q \in \mathbb{F}_2[x]$ be monic of degree $n - 1$ with $q(0) \neq 0$. Then there is a similarity class D of $\mathrm{GL}_n(\mathbb{F}_2)$ whose minimal polynomial is divisible by q and such that $M \in DX$.*

We do not prove these two theorems. Every other step is elementary, and every other step is checked in Lean 4 [7] with Mathlib [8]: the formal main theorem takes Theorems A and B as hypotheses and depends on no other assumptions (Section 8).

1.5. **How the result was found.** This work was produced almost entirely by an AI agent (Droid, by Factory), working under the author's direction. The author is developing a new computational mathematics project, a programming language for exact mathematical computation in which one writes definitions as on paper and states expected facts as checks that the program evaluates, and this problem was taken up to test that project on open problems. The agent carried out the computations, found the argument, searched the literature, wrote

the Lean formalization and drafted this text. The author chose the problem, set the scope (in particular, to cite Theorems A and B rather than formalize them) and takes responsibility for the paper. No specialist has refereed the argument; the main evidence for its correctness is the Lean check of every step outside Theorems A and B.

The path to the proof went as follows.

Small cases. An exhaustive computation in $\mathrm{GL}_3(\mathbb{F}_2)$ and $\mathrm{GL}_4(\mathbb{F}_2)$ confirmed the answer for $n = 3, 4$ and produced the data of Table 1. Random searches for explicit products found witnesses for every pair of classes for $n = 5$ and $n = 6$. This suggested that the answer is yes and that most classes have a lot of room to spare.

Matrices with a J1 block. If $\alpha = 1 \oplus \alpha'$, induction handles every target g that also has a J1 block. The data suggested that, for every other target, one conjugate c of α can be chosen so that gc^{-1} has a J1 block; this became Lemma 5.2. Testing a construction of c on every pair of classes for $n = 4, 5, 6$ succeeded outside one situation: α an involution and g satisfying $g^2 + g + 1 = 0$. In $\mathrm{GL}_4(\mathbb{F}_2)$, none of the 315 involutions works for such g. Remark 5.3 explains why, and Lemma 5.4 handles this case directly.

Matrices without a J1 block. Here the reduction does not apply, and several attempts at a direct argument stalled. A literature search found Theorems A and B, which together turn "α has large minimum rank" into "every g is a short product of conjugates". What remained was to show that matrices without a J1 block have minimum rank at least 3 (Lemma 6.1).

Formalization. The first complete argument split the classes by minimum rank and used a third published result for transvections. Writing the proof in Lean led to the simpler split used here, by whether α has a J1 block, and the third result was no longer needed.

1.6. **Organization.** Section 2 explains the idea of the proof. Section 3 collects preliminaries, and Section 4 treats $n = 3, 4$ and gives the exact data. Sections 5 and 6 treat matrices with and without a J1 block, and Section 7 completes the proofs. Section 8 describes the formalization and Section 9 lists open questions.

2. The idea of the proof

Fix $\alpha \neq 1$ and write C for its class. Two simple facts organize everything.

First, adding factors never hurts: once every nonidentity element lies in C^m, the same holds for C^k with $k \geq m$ (Lemma 3.1). So it is enough to reach some $m \leq n$.

Second, matrices come in two kinds. A matrix with a J1 block is "the identity plus something smaller". A matrix without one is spread out: no eigenvalue can have a large eigenspace.

If α has a J1 block, write $\alpha = 1 \oplus \alpha'$. A target of the same shape, $g = 1 \oplus g'$, is handled in dimension $n - 1$ by induction. For any other target g, we choose one conjugate c of α so that gc^{-1} has a J1 block. Then gc^{-1} is a product of $n - 1$ conjugates by the previous case, and $g = (gc^{-1})c$ is a product of n. Finding c is a matter of lining up two pairs of vectors and linear forms, one adapted to g and one to α. This works unless α is an involution and g is a sum of rotations of order 3, and that case is done by hand, using dimension $n - 2$.

If α has no J1 block, its minimum rank is at least 3. Theorem A then says that a product of about $(n - 1)/3$ conjugates of α is a cyclic matrix D. Theorem B says that every non-homothety factors as (a matrix of a special class N) times (a conjugate of D). The class N is the same whatever matrix is factored, and applying Theorem B to α itself shows that $N \subseteq C \cdot C^k$. So every g lies in $N \cdot C^k \subseteq C^{2k+1}$, and $2k + 1 \leq n$.

The base cases $n = 3$ and $n = 4$ are done by computer, because the induction needs dimensions $n - 1$ and $n - 2$ and Theorem A excludes $n = 4$.

3. Preliminaries

Throughout, V is a vector space of dimension n over $\mathbb{F}_2$, V^* is its dual and $G = \mathrm{GL}(V)$. For $a \in G$ let C_a be its conjugacy class and let C_a^k be the set of products of k elements of C_a. The set C_a^k is closed under conjugation and depends only on C_a. For $n \geq 2$ the centre of G is trivial, so every $a \neq 1$ is non-central.

For $n \geq 1$ let $P(n)$ be the statement: for every space V of dimension n and all $a, g \in \mathrm{GL}(V) \setminus \{1\}$, we have $g \in C_a^k$ for every $k \geq n$.

Lemma 3.1. *Let $a \in G$ be non-central and suppose $G \setminus \{1\} \subseteq C_a^m$. Then $G \setminus \{1\} \subseteq C_a^k$ for every $k \geq m$.*

Proof. Let $k = m + j$ with $j \geq 1$ and $g \neq 1$. The set C_a^j contains a^j and $(xax^{-1})a^{j-1}$, where $xax^{-1} \neq a$, and these differ. So some $u \in C_a^j$ satisfies $u \neq g$. Then $u^{-1}g \neq 1$ lies in C_a^m, and $g = u \cdot u^{-1}g \in C_a^{j+m}$. □

Over $\mathbb{F}_2$, vectors $x_1, \ldots, x_m$ are linearly independent if and only if no nonempty subset of them sums to 0. A vector space over any field is not the union of two proper subspaces. If $x_1, \ldots, x_m \in V$ are linearly independent, there is $\varphi \in V^*$ with any prescribed values $\varphi(x_i)$.

Definition 3.2. An element $h \in G$ has a *J1 block* if there are $v \in V$ and $\varphi \in V^*$ with $\varphi(v) = 1$, $hv = v$ and $\varphi \circ h = \varphi$.

If h has a J1 block then $V = \langle v \rangle \oplus \ker \varphi$, both summands are h-invariant, and $h = 1 \oplus h'$ with $h' \in \mathrm{GL}(\ker \varphi)$. Having a J1 block is invariant under conjugation. With $T = h - 1$, a J1 block exists exactly when $\ker T \not\subseteq \operatorname{im} T$, that is, when $n - 2\operatorname{rank} T + \operatorname{rank} T^2 > 0$.

A *frame* of size m is a pair of families $e_1, \ldots, e_m \in V$ and $f_1, \ldots, f_m \in V^*$ with $f_i(e_j) = \delta_{ij}$. Then $V = \operatorname{span}(e_1, \ldots, e_m) \oplus \bigcap_i \ker f_i$.

Lemma 3.3. *Given two frames (e, f) and (e', f') of the same size m, there is $t \in G$ with $te'_j = e_j$ and $f_i \circ t = f'_i$ for all i, j.*

Proof. Both kernels $\bigcap_i \ker f_i$ and $\bigcap_i \ker f'_i$ have dimension $n - m$. Let t send e'_j to e_j and map the second kernel isomorphically onto the first. For $x = \sum_j f'_j(x)e'_j + y$ with y in the second kernel, $tx = \sum_j f'_j(x)e_j + ty$, so $f_i(tx) = f'_i(x)$. □

We also use the standard fact that if $a = a_1 \oplus a_2$ along $V = A \oplus B$ and $b = b_1 \oplus b_2$ along $V = A' \oplus B'$, where a_1 is conjugate to b_1 by an isomorphism $A \to A'$ and a_2 to b_2 by an isomorphism $B \to B'$, then a and b are conjugate in G. Products of conjugates in $\mathrm{GL}(A) \times \mathrm{GL}(B)$ can be formed in each factor separately.

4. The cases $n = 3$ and $n = 4$

The group $\mathrm{GL}_3(\mathbb{F}_2)$ has order 168 and 6 conjugacy classes, and $\mathrm{GL}_4(\mathbb{F}_2)$, which is isomorphic to the alternating group A_8, has order 20160 and 14 conjugacy classes.

The proof. We fix one representative of each class. A certificate consists of, for each invertible matrix, a matrix conjugating it to its representative, and, for each ordered pair (a, b) of nonidentity representatives, n explicit conjugates of a whose product is b. The certificates were found by a computer search. Lean checks them by evaluation in its kernel; the search itself is not part of the proof. This shows that every nonidentity element is a product of n conjugates of every nonidentity element, and Lemma 3.1 gives $P(3)$ and $P(4)$.

The exact numbers. For $\alpha \neq 1$ let $\mathrm{cn}(\alpha)$ be the least m such that every nonidentity element lies in C_α^m. By Lemma 3.1, every nonidentity element then lies in C_α^k for all $k \geq \mathrm{cn}(\alpha)$.

Theorem 1 says $\mathrm{cn}(\alpha) \le n$. Table 1 lists $\mathrm{cn}(\alpha)$ for every class of $\mathrm{GL}_3(\mathbb{F}_2)$ and $\mathrm{GL}_4(\mathbb{F}_2)$. It was computed exhaustively by a separate program, working with whole classes: since C_α^k is a union of classes, C_α^{k+1} is found by multiplying one representative of each class in C_α^k by every element of C_α. This computation is not formalized, but it agrees with the Lean-checked bound $\mathrm{cn}(\alpha) \le n$.

In the table, J_k is the unipotent Jordan block of size k, ω is the companion matrix of x^2+x+1, and $C(p)$ is the companion matrix of p.

class of α	order	size	J1 block	$\delta(\alpha)$	$\mathrm{cn}(\alpha)$
$\mathrm{GL}_3(\mathbb{F}_2)$					
$J_2 \oplus 1$ (transvection)	2	21	yes	1	3
$\omega \oplus 1$	3	56	yes	2	2
J_3	4	42	no	2	2
$C(x^3+x+1)$	7	24	no	2	3
$C(x^3+x^2+1)$	7	24	no	2	3
$\mathrm{GL}_4(\mathbb{F}_2)$					
$J_2 \oplus 1 \oplus 1$ (transvection)	2	105	yes	1	4
$J_2 \oplus J_2$	2	210	no	2	3
$\omega \oplus 1 \oplus 1$	3	1120	yes	2	2
$\omega \oplus \omega$	3	112	no	2	4
$J_3 \oplus 1$	4	1260	yes	2	2
J_4	4	2520	no	3	2
$C(x^4+x^3+x^2+x+1)$	5	1344	no	3	3
$J_2 \oplus \omega$	6	3360	no	3	2
$C\big((x^2+x+1)^2\big)$	6	1680	no	3	2
$C(x^3+x+1) \oplus 1$	7	2880	yes	3	2
$C(x^3+x^2+1) \oplus 1$	7	2880	yes	3	2
$C(x^4+x+1)$	15	1344	no	3	2
$C(x^4+x^3+1)$	15	1344	no	3	2

TABLE 1. The least m such that every nonidentity element is a product of m conjugates of α, for every nonidentity class of $\mathrm{GL}_3(\mathbb{F}_2)$ and $\mathrm{GL}_4(\mathbb{F}_2)$.

Three features of the table are worth noting.

- The bound n is attained by the transvections, as the rank argument of the introduction predicts, and in $\mathrm{GL}_4(\mathbb{F}_2)$ also by $\omega \oplus \omega$. This is the fixed-point-free element of order 3, the same kind of matrix ($g^2+g+1=0$) that is the exceptional target in Lemma 5.2.
- Most classes need only 2 conjugates, far below n.
- In $\mathrm{GL}_4(\mathbb{F}_2)$ the classes $J_2 \oplus J_2$ and $\omega \oplus \omega$ have no J1 block but minimum rank 2. So Lemma 6.1 below fails for $n=4$, and its hypothesis $n \ge 5$ is needed.

5. Matrices with a J1 block

In this section $n \ge 5$, and we assume $P(n-1)$ and $P(n-2)$.

Lemma 5.1. *If $\alpha \ne 1$ and $h \ne 1$ both have a J1 block, then $h \in C_\alpha^k$ for every $k \ge n-1$.*

Proof. Write $\alpha = 1 \oplus \alpha'$ and $h = 1 \oplus h'$ along splittings $\langle v \rangle \oplus U$ and $\langle w \rangle \oplus U'$. Then $\alpha' \ne 1$ and $h' \ne 1$. After conjugating α we may assume the two splittings coincide. By $P(n-1)$, h'

is a product of k conjugates of α' in $\mathrm{GL}(U')$, and adding the identity on the first summand gives $h \in C_\alpha^k$. □

Lemma 5.2 (Lemma P)**.** *Let $\alpha \neq 1$ have a J1 block and let $g \neq 1$. If $\alpha^2 = 1$, assume also $g^2 + g + 1 \neq 0$. Then there is $c \in C_\alpha$ such that gc^{-1} has a J1 block.*

Proof. Step 1. Suppose $v, w \in V$ and $\varphi, \varphi' \in V^*$ satisfy, with $a = \varphi(g^{-1}v)$,

$$\varphi(v) = 1, \quad \varphi(gv) = 0, \quad \varphi'(w) = a, \quad \varphi'(\alpha w) = 1, \quad \varphi'(\alpha^2 w) = 0. \tag{1}$$

Then the vectors $g^{-1}v$, v with the covectors $\varphi \circ g$, $\varphi - a\,\varphi \circ g$ form a frame, and so do w, αw with $\varphi' \circ \alpha$, $\varphi' - a\,\varphi' \circ \alpha$. Let t be given by Lemma 3.3, so that $tw = g^{-1}v$, $t\alpha w = v$, $\varphi \circ g \circ t = \varphi' \circ \alpha$ and $(\varphi - a\,\varphi \circ g) \circ t = \varphi' - a\,\varphi' \circ \alpha$. The last two give $\varphi \circ t = \varphi'$. Put $c = t\alpha t^{-1}$ and $h = gc^{-1} = gt\alpha^{-1}t^{-1}$. Then

$$hv = gt\alpha^{-1}(\alpha w) = gtw = v, \qquad \varphi(hx) = (\varphi \circ g \circ t)(\alpha^{-1}t^{-1}x) = \varphi'(t^{-1}x) = \varphi(x),$$

so h has the J1 block (v, φ).

Step 2: $\alpha^2 \neq 1$. Pick v with $gv \neq v$. Then v and gv are linearly independent, so there is φ with $\varphi(v) = 1$ and $\varphi(gv) = 0$. Let (u, ψ) be a J1 block of α. Then $(\alpha^2+\alpha+1)u = 3u = u \neq 0$, so both $\ker(\alpha^2 + \alpha + 1)$ and $\ker(\alpha^2 - 1)$ are proper subspaces, and some w lies in neither. The vectors $w, \alpha w, \alpha^2 w$ are linearly independent: none is 0, they are pairwise distinct because $\alpha^2 w \neq w$, and their sum is not 0. So there is φ' with the values required in (1).

Step 3: $\alpha^2 = 1$ *and* $g^2+g+1 \neq 0$. Since $g \neq 1$, the subspaces $\ker(g-1)$ and $\ker(g^2+g+1)$ are proper, so some v lies in neither. Then $v \notin \mathrm{span}(gv, g^{-1}v)$: indeed $v \neq 0$, $v \neq gv$, $v \neq g^{-1}v$, and $v \neq gv + g^{-1}v$ because applying g would give $(g^2+g+1)v = 0$. Choose φ with $\varphi(v) = 1$ and $\varphi(gv) = \varphi(g^{-1}v) = 0$, so $a = 0$. Pick w with $\alpha w \neq w$ and φ' with $\varphi'(w) = 0$ and $\varphi'(\alpha w) = 1$. Then $\varphi'(\alpha^2 w) = \varphi'(w) = 0$, and (1) holds. □

Remark 5.3. The extra hypothesis in Lemma 5.2 cannot be dropped. Let $\alpha^2 = 1$ and $g^2+g+1 = 0$, and suppose $c \in C_\alpha$ and $h = gc^{-1}$ has a J1 block (v, φ). Since $c = c^{-1}$, the equation $hv = v$ gives $cv = g^{-1}v$, and $\varphi \circ h = \varphi$ gives $\varphi \circ g = \varphi \circ c$. Evaluating at v gives $\varphi(gv) = \varphi(g^{-1}v)$. But $g(g+1) = g^2 + g = 1$, so $g^{-1} = g+1$ and $\varphi(g^{-1}v) = \varphi(gv) + \varphi(v) = \varphi(gv) + 1$, a contradiction.

The case excluded in Lemma 5.2 is handled directly. In $\mathrm{GL}_2(\mathbb{F}_2)$, let σ be the swap of the two coordinates and $\omega = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, of order 3. Then $\omega = \sigma \cdot (\omega\sigma\omega^{-1})$ is a product of two conjugates of σ, and since $\sigma^2 = 1$ it is a product of $2j$ conjugates of σ for every $j \geq 1$.

Lemma 5.4. *Let $\alpha \neq 1$ with $\alpha^2 = 1$, and let g satisfy $g^2 + g + 1 = 0$. Then n is even and $g \in C_\alpha^n$.*

Proof. For $v \neq 0$ we have $gv \neq v$, since otherwise $v = 3v = (g^2 + g + 1)v = 0$. Choose ψ with $\psi(v) = 1$ and $\psi(gv) = 0$. Using $g^2 = g + 1$, the pairs v, gv and $\psi, \psi \circ g$ form a frame, and both $\mathrm{span}(v, gv)$ and $\ker\psi \cap \ker(\psi \circ g)$ are g-invariant. In the basis v, gv the map g acts as ω, so $g = \omega \oplus g'$, where g' again satisfies $g'^2 + g' + 1 = 0$. By induction n is even. In the same way, for u with $\alpha u \neq u$, the plane $\mathrm{span}(u, \alpha u)$ and a complement are α-invariant and $\alpha = \sigma \oplus \alpha'$. After conjugating α, both decompositions use the same splitting $V = P \oplus U$ with $\dim U = n - 2 \geq 3$. The map g' is not the identity.

If $\alpha' \neq 1$, then $g' \in C_{\alpha'}^n$ by $P(n-2)$ and $\omega \in C_\sigma^n$ because n is even, so $g = \omega \oplus g' \in C_\alpha^n$.

If $\alpha' = 1$, write $g = (\omega \oplus 1)(1 \oplus g')$. The first factor lies in C_α^2. Split $U = P' \oplus U'$ with $g' = \omega \oplus g''$ as above and let $\tau = \sigma \oplus 1$ on U. Then $\tau \neq 1$, so $P(n-2)$ gives $g' \in C_\tau^{n-2}$. The map $1_P \oplus \tau$ is conjugate to α, since both act as σ on a plane and as the identity on a complement. Hence $1 \oplus g' \in C_\alpha^{n-2}$ and $g \in C_\alpha^n$. □

Proposition 5.5. *Let $n \geq 5$ and assume $P(n-1)$ and $P(n-2)$. If $\alpha \neq 1$ has a J1 block, then every $g \neq 1$ lies in C_α^n.*

Proof. If g has a J1 block, use Lemma 5.1. If $\alpha^2 = 1$ and $g^2 + g + 1 = 0$, use Lemma 5.4. Otherwise Lemma 5.2 gives $c \in C_\alpha$ such that $h = gc^{-1}$ has a J1 block. Then $h \neq 1$, since otherwise $g = c$ would have a J1 block. By Lemma 5.1, $h \in C_\alpha^{n-1}$, and $g = hc \in C_\alpha^n$. □

6. Matrices without a J1 block

Lemma 6.1. *Let $n \geq 5$ and let $\alpha \in \mathrm{GL}_n(\mathbb{F}_2)$ have no J1 block. Then $\delta(\alpha) \geq 3$.*

Proof. Let L be an algebraic closure of $\mathbb{F}_2$ and $\lambda \in L$. For $\lambda = 0$ the rank of $\alpha - \lambda$ is n.

Let $\lambda = 1$ and suppose $\mathrm{rank}(\alpha - 1) \leq 2$. The fixed space $\ker(\alpha - 1)$ has dimension at least $n - 2 \geq 3$, so it is not contained in the image of $\alpha - 1$. Take a fixed vector v outside the image and $\varphi \in V^*$ with $\varphi(v) = 1$ that vanishes on the image. Then $\varphi \circ \alpha = \varphi$, and (v, φ) is a J1 block, a contradiction.

Let $\lambda \notin \{0, 1\}$ and let $E_\mu \subseteq L^n$ be the μ-eigenspace of α. Since the entries of α lie in $\mathbb{F}_2$, squaring every coordinate maps E_λ into E_{λ^2}. This map is injective and semilinear with respect to the Frobenius automorphism of L, so $\dim E_\lambda \leq \dim E_{\lambda^2}$. As $\lambda^2 \neq \lambda$, the two eigenspaces intersect in 0, so $2 \dim E_\lambda \leq n$ and $\mathrm{rank}(\alpha - \lambda) \geq n - \lfloor n/2 \rfloor \geq 3$. □

Lemma 6.2. *Let $n \geq 3$ and let $r \in \mathbb{F}_2[x]$ be irreducible of degree $n - 1$. If $u \in \mathrm{GL}_n(\mathbb{F}_2)$ has minimal polynomial divisible by r, then its minimal and characteristic polynomials both equal $r \cdot (x + 1)$. In particular any two such matrices are similar.*

Proof. The characteristic polynomial χ_u is divisible by r, so $\chi_u = r\,s$ with s monic of degree 1. Since $\det u \neq 0$, $s(0) \neq 0$, so $s = x + 1$. The minimal polynomial is r or $r(x+1)$. Since 1 is a root of χ_u, it is an eigenvalue of u and a root of the minimal polynomial, while r, irreducible of degree at least 2, has no root in $\mathbb{F}_2$. So the minimal polynomial is $r(x + 1) = \chi_u$, and u is similar to the companion matrix of $r(x + 1)$. □

Proposition 6.3. *Assume Theorems A and B. Let $n \geq 5$, let $\alpha \in \mathrm{GL}_n(\mathbb{F}_2)$, $\alpha \neq 1$, have no J1 block, and let $k = \lceil (n-1)/3 \rceil$. Then every $g \neq 1$ lies in C_α^{2k+1}.*

Proof. Write $C = C_\alpha$. By Lemma 6.1, k copies of C have minimum ranks summing to at least $3k \geq n - 1$, and C is non-central. Theorem A gives a cyclic matrix $D \in C^k$. Its inverse is cyclic too: if $v, Dv, \ldots, D^{n-1}v$ is a basis, so is the sequence $D^{-j}(D^{n-1}v)$, $0 \leq j < n$.

Let r be the minimal polynomial over $\mathbb{F}_2$ of a primitive element of $\mathbb{F}_{2^{n-1}}$. It is monic and irreducible of degree $n - 1 \geq 2$, and $r(0) \neq 0$. The only homothety in $\mathrm{GL}_n(\mathbb{F}_2)$ is 1, so α and g are not homotheties.

Theorem B with $M = \alpha$ and X the class of D^{-1} gives $\alpha = d'x'$ with x' similar to D^{-1} and r dividing the minimal polynomial of d'. Then x'^{-1} is similar to D, so $x'^{-1} \in C^k$ and $d' = \alpha x'^{-1} \in C^{k+1}$.

Theorem B with $M = g$ and X the class of D gives $g = dx$ with x similar to D, so $x \in C^k$, and r dividing the minimal polynomial of d. By Lemma 6.2, d is similar to d', so $d \in C^{k+1}$ and $g = dx \in C^{2k+1}$. □

7. Proof of the theorems

Proof of Theorem 1. We prove $P(n)$ for all $n \geq 3$ by strong induction. The cases $n = 3$ and $n = 4$ are Section 4. Let $n \geq 5$ and $\alpha \neq 1$. If α has a J1 block, Proposition 5.5 (which uses $P(n-1)$ and $P(n-2)$, with $n - 2 \geq 3$) gives $G \setminus \{1\} \subseteq C_\alpha^n$. Otherwise Proposition 6.3 gives

$G \setminus \{1\} \subseteq C_\alpha^{2k+1}$ with $k = \lceil (n-1)/3 \rceil$, and $2k+1 \le n$ for $n \ge 5$. In both cases Lemma 3.1 gives $G \setminus \{1\} \subseteq C_\alpha^{k'}$ for every $k' \ge n$. □

Proof of Theorem 2. Combine Proposition 6.3 with Lemma 3.1. □

Corollary 7.1. *Assume Theorems A and B. For $n \ge 3$ and $\alpha \in \mathrm{GL}_n(\mathbb{F}_2)$, $\alpha \ne 1$, the identity is a product of $n+1$ conjugates of α.*

Proof. By Theorem 1, $\alpha^{-1} = c_1 \cdots c_n$ with $c_i \in C_\alpha$, so $1 = c_1 \cdots c_n \alpha$. □

For a single class repeated $n+1$ times, this is the $\mathbb{F}_2$ analogue of the statement that the extended covering number is at most $n+1$, which Knüppel and Nielsen prove for arbitrary classes over fields that are not too small.

8. The formalization

The Lean development accompanies this note. It consists of about 2300 lines of Lean together with generated certificate data, and builds against a pinned version of Mathlib. Its main theorem is

```
theorem Vaserstein.question :
    KnuppelNielsen69 -> BungerNielsen12 -> Question (ZMod 2)
```

together with `Vaserstein.questionFrom`, the version with $k \ge n$ factors.

What is stated. `Question F` is defined directly from the wording of Problem 16.21, with `SpecialLinearGroup`, its centre, and similarity by an invertible matrix. It quantifies over all $n > 2$ and all non-central α, β. `KnuppelNielsen69` and `BungerNielsen12` are the statements of Theorems A and B for $\mathbb{F}_2$, with cyclicity and minimum rank defined as in Section 1. The minimum rank is defined through an algebraic closure, as in the formula of Knüppel and Nielsen.

What is proved. Everything else in this note, including the cases $n = 3, 4$, which are proved with no hypotheses. Their certificates, about 20,000 conjugating matrices for $n = 4$, are checked with `decide +kernel`, so the check runs in Lean's kernel and not in compiled code. The development uses no `sorry`, no `native_decide` and no axioms beyond Lean's standard `propext`, `Classical.choice` and `Quot.sound`.

What must be trusted. Lean's kernel and Mathlib; the published proofs of Theorems A and B; and the transcription of those two theorems into Lean, which was compared by hand with the statements in the papers. An error in the transcription would make the Lean theorem a statement about slightly different hypotheses, so this is the step a careful reader should check. The exact numbers of Table 1 come from a separate computation and are not formalized.

9. Discussion and open questions

Question 9.1. Can Theorems A and B be formalized, so that the answer for $\mathbb{F}_2$ becomes unconditional? Theorem B concerns a single field and an explicit construction and seems the more accessible of the two.

Question 9.2. Does the same argument work over other small fields? The reduction for matrices with a J1 block uses $\mathbb{F}_2$ in several places, for instance the test for linear independence by sums of subsets, the identity $3u = u$ in Lemma 5.2, and the structure of $\mathrm{GL}_2(\mathbb{F}_2)$ in Lemma 5.4.

Question 9.3. What is $\mathrm{cn}(\alpha)$ for general n? Table 1 and Theorem 2 suggest that $\mathrm{cn}(\alpha)$ is much smaller than n unless α is close to the identity or to a sum of order-3 rotations. Is $\mathrm{cn}(\alpha)$ bounded by a constant for cyclic α, as Lev's results give, under conditions on the eigenvalues, for larger fields?

Question 9.4. Can the special cases used here be proved directly? Only the case of Theorem A with all classes equal is used, and Theorem B is used for a single polynomial q in each dimension.

Use of AI. As described in Section 1, this work was produced almost entirely by an AI agent (Droid, by Factory) under the author's direction, while testing a new computational mathematics project. The agent carried out the computations, found the argument, searched the literature, wrote the Lean formalization and drafted this text; the author chose the problem, set the scope and takes responsibility for the paper. The Lean check covers every step except Theorems A and B, but it does not cover the exposition, the literature survey or the claims about prior work, which have not been reviewed by a specialist. Corrections are welcome.

References

[1] Z. Arad and M. Herzog (eds.), *Products of Conjugacy Classes in Groups*, Lecture Notes in Math. **1112**, Springer, 1985.

[2] F. Bünger and K. Nielsen, *A matrix-decomposition theorem for* $\mathrm{GL}_n(K)$, Linear Algebra Appl. **298** (1999), 39–50, doi:10.1016/S0024-3795(99)00138-X.

[3] F. Knüppel, *Products of similar matrices*, Linear Algebra Appl. **233** (1996), 99–108.

[4] F. Knüppel and K. Nielsen, *The extended covering number of* SL_n *is* $n+1$, Linear Algebra Appl. **418** (2006), 634–656, doi:10.1016/j.laa.2006.03.011.

[5] E. I. Khukhro and V. D. Mazurov (eds.), *Unsolved Problems in Group Theory. The Kourovka Notebook*, No. 21, 2026, arXiv:1401.0300.

[6] A. Lev, *Products of cyclic similarity classes in the groups* $\mathrm{GL}_n(F)$, Linear Algebra Appl. **202** (1994), 235–266.

[7] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction, CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[8] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), 367–381.